

```

sysadmin@localhost:~$ tar -cvf mybackups/udev.tar /etc/udev
tar: Removing leading `/' from member names
/etc/udev/
/etc/udev/hwdb.d/
/etc/udev/rules.d/
/etc/udev/rules.d/70-persistent-cd.rules
/etc/udev/rules.d/README
/etc/udev/udev.conf
sysadmin@localhost:~$

```

The `tar` command is used to merge multiple files into a single file. By default, it does not compress the data.

```

sysadmin@localhost:~$ tar -tvf mybackups/udev.tar
drwxr-xr-x root/root      0 2021-02-08 15:07 etc/udev/
drwxr-xr-x root/root      0 2021-01-06 21:04 etc/udev/hwdb.d/
drwxr-xr-x root/root      0 2021-02-08 15:08 etc/udev/rules.d/
-rw-r--r-- root/root    306 2021-02-08 15:08 etc/udev/rules.d/70-persistent-c
d.rules
-rw-r--r-- root/root   1157 2021-02-08 15:08 etc/udev/rules.d/README
-rw-r--r-- root/root    218 2021-02-08 15:08 etc/udev/udev.conf
sysadmin@localhost:~$

```

Display the contents of a `tar` file by using the available options (`t` = list contents, `v` = verbose, `f` = filename):

```

sysadmin@localhost:~$ tar -zcvf mybackups/udev.tar.gz /etc/udev
tar: Removing leading `/' from member names
/etc/udev/
/etc/udev/hwdb.d/
/etc/udev/rules.d/
/etc/udev/rules.d/70-persistent-cd.rules
/etc/udev/rules.d/README
/etc/udev/udev.conf
sysadmin@localhost:~$

```

To create a `tar` file that is compressed, use the `-z` option. The `-z` option makes use of the `gzip` utility to perform compression.

```

sysadmin@localhost:~$ cd mybackups
sysadmin@localhost:~/mybackups$ ls
udev.tar  udev.tar.gz
sysadmin@localhost:~/mybackups$ tar -xvf udev.tar.gz
etc/udev/
etc/udev/hwdb.d/
etc/udev/rules.d/
etc/udev/rules.d/70-persistent-cd.rules
etc/udev/rules.d/README
etc/udev/udev.conf
sysadmin@localhost:~/mybackups$ ls
etc  udev.tar  udev.tar.gz
sysadmin@localhost:~/mybackups$ ls etc
udev
sysadmin@localhost:~/mybackups$ ls etc/udev
hwdb.d  rules.d  udev.conf
sysadmin@localhost:~/mybackups$ ls etc/udev/rules.d
70-persistent-cd.rules  README
sysadmin@localhost:~/mybackups$

```

Extract the contents of an archive. Data is restored to the current directory by default.

```

sysadmin@localhost:~$ cd mybackups
sysadmin@localhost:~/mybackups$ tar -rvf udev.tar /etc/hosts
tar: Removing leading `/' from member names
/etc/hosts
sysadmin@localhost:~/mybackups$ tar -rvf udev.tar
sysadmin@localhost:~/mybackups$ tar -tvf udev.tar
drwxr-xr-x root/root      0 2021-02-08 15:07 etc/udev/
drwxr-xr-x root/root      0 2021-01-06 21:04 etc/udev/hwdb.d/
drwxr-xr-x root/root      0 2021-02-08 15:08 etc/udev/rules.d/
-rw-r--r-- root/root    306 2021-02-08 15:08 etc/udev/rules.d/70-persistent-c
d.rules
-rw-r--r-- root/root    1157 2021-02-08 15:08 etc/udev/rules.d/README
-rw-r--r-- root/root     218 2021-02-08 15:08 etc/udev/udev.conf
-rw-r--r-- root/root     172 2026-10-04 17:42 etc/hosts
-rw-r--r-- root/root     172 2026-10-04 17:42 etc/hosts
sysadmin@localhost:~/mybackups$

```

To add a file to an existing archive, use the `-r` option to the `tar` command. Execute the following commands to perform this action and verify the existence of the new file in the `tar` archive:

```
sysadmin@localhost:~/mybackups$ cp /usr/share/dict/words .
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct  4 18:07 words
sysadmin@localhost:~/mybackups$ gzip words
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct  4 18:07 words.gz
sysadmin@localhost:~/mybackups$
```

Using the commands to compress a copy of the `words` file:

```
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct  4 18:07 words.gz
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct  4 18:07 words.gz
sysadmin@localhost:~/mybackups$ gunzip words.gz
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct  4 18:07 words
sysadmin@localhost:~/mybackups$
```

Using the commands to uncompress the `words.gz` file:

```
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct  4 18:07 words
sysadmin@localhost:~/mybackups$ bzip2 words
sysadmin@localhost:~/mybackups$ ls -l words.bz2
-rw-r--r-- 1 sysadmin sysadmin 345560 Oct  4 18:07 words.bz2
sysadmin@localhost:~/mybackups$
```

Using `bzip2` and `bunzip2` to compress and uncompress a file is very similar to using `gzip` and `gunzip`. The compressed file is created with a `.bz2` extension. The extension is removed when uncompressed. Execute the following commands to compress a copy of the `words` file:

```
sysadmin@localhost:~/mybackups$ ls -l words.bz2
-rw-r--r-- 1 sysadmin sysadmin 345560 Oct  4 18:07 words.bz2
sysadmin@localhost:~/mybackups$ bunzip2 words.bz2
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct  4 18:07 words
sysadmin@localhost:~/mybackups$
```

Using `bzip2` and `bunzip2` to compress and uncompress a file is very similar to using `gzip` and `gunzip`. The compressed file is created with a `.bz2` extension. The extension is removed when uncompressed. Execute the following commands to compress a copy of the `words` file:

```
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct  4 18:07 words
sysadmin@localhost:~/mybackups$ xz words
sysadmin@localhost:~/mybackups$ ls -l words.xz
-rw-r--r-- 1 sysadmin sysadmin 198756 Oct  4 18:07 words.xz
sysadmin@localhost:~/mybackups$
```

Using `xz` and `unxz` to compress and uncompress a file is also very similar to using `gzip` and `gunzip`. The compressed file is created with a `.xz` extension. The extension is removed when uncompressed. Execute the following commands to compress a copy of the `words` file: